Supplement: A reduced Kirchhoff-metric variational integrator for multiple ellipsoids in potential flow

Abstract

This supplement develops an alternative to the impulse scheme of the main document. Because the fluid in potential flow carries no independent state, the coupled N-body system is exactly a finite-dimensional Lagrangian system whose kinetic energy is a configuration-dependent quadratic form in the body velocities—the Kirchhoff (added-mass) metric $\mathsf{M}(q)$. We show how to assemble $\mathsf{M}(q)$ explicitly and cheaply from the same boundary-element (BEM) solve used elsewhere, cast the dynamics as the body-frame Kirchhoff equations, and integrate them with a structure-preserving Lie-group scheme: an implicit-midpoint update whose momentum is transported by the exact SE(3) coadjoint action. The resulting integrator is *fully determined* (there is no per-body impulse ambiguity), conserves the total linear and angular momentum maps to machine precision, keeps the energy bounded with no secular drift, and is second-order accurate. A predictor form reduces the cost to a small multiple of the impulse scheme's per-step cost.

1 The fluid as a configuration-space metric

Collect the rigid-body configuration as $q = \{\mathbf{c}_i, R_i\}_{i=1}^N$ with $R_i = R(q_i) \in \mathrm{SO}(3)$ the rotation carried by the unit quaternion q_i, and stack the lab-frame generalized velocity

$$\mathbf{z} = (\mathbf{v}_1, \boldsymbol{\omega}_1, \dots, \mathbf{v}_N, \boldsymbol{\omega}_N) \in \mathbb{R}^{6N}. \tag{1}$$

As emphasised in the main document, eqs. (2)–(3) contain no time derivative of ϕ: the potential—and hence every load—is an instantaneous linear functional of $\mathbf{z}$ at fixed geometry. In particular the stacked Lamb impulses of eqs. (8)–(9),

$$\boldsymbol{\ell}(q, \mathbf{z}) = (\mathbf{L}_1, \boldsymbol{\Lambda}_1, \dots, \mathbf{L}_N, \boldsymbol{\Lambda}_N), \tag{2}$$

are linear in $\mathbf{z}$ and vanish at $\mathbf{z} = \mathbf{0}$, so they define a symmetric *added-mass matrix*

$$\boxed{\boldsymbol{\ell}(q, \mathbf{z}) = -\,\mathsf{M}_a(q)\,\mathbf{z}, \qquad \mathsf{M}_a(q) = -\,\frac{\partial \boldsymbol{\ell}}{\partial \mathbf{z}}} \tag{3}$$

The sign is that of the main document: for a single translating body $\mathbf{L}_i = -\mathsf{M}_f \mathbf{v}_i$, so that $\mathbf{p}_i = m_i \mathbf{v}_i - \mathbf{L}_i$ (eq. (12)) carries the effective inertia $m_i + \mathsf{M}_f$. The diagonal blocks of M_a are the (configuration-dependent) self added masses—reducing, for an isolated body, to the analytic tensors of §4—and the off-diagonal blocks are the body–body hydrodynamic coupling. The fluid kinetic energy of eq. (10) is the associated quadratic form,

$$T_f(q, \mathbf{z}) = \tfrac{1}{2}\,\mathbf{z}^\top \mathsf{M}_a(q)\,\mathbf{z}. \tag{4}$$

Adding the solid kinetic energy T_s gives the reduced Lagrangian

$$\mathcal{L}(q, \mathbf{z}) = \tfrac{1}{2}\,\mathbf{z}^\top \mathsf{M}(q)\,\mathbf{z}, \qquad \mathsf{M}(q) = \mathsf{M}_s(q) + \mathsf{M}_a(q), \tag{5}$$

with the block-diagonal solid metric $\mathsf{M}_s = \mathrm{blkdiag}\big(m_i \mathsf{I}_3,\ R_i I_i^{\mathrm{body}} R_i^\top\big)$. Equation (5) is the Kirchhoff/Lamb reduction: the coupled motion is geodesic motion in the metric $\mathsf{M}(q)$ on the configuration manifold $Q = \mathrm{SE}(3)^N$.

Why this matters. The generalized momentum $\boldsymbol{\mu} = \mathsf{M}(q)\mathbf{z}$ is a single global covector, and the equations of motion involve one well-defined configuration force $\frac{1}{2}\,\partial_q(\mathbf{z}^\top \mathsf{M}\mathbf{z})$. There is no per-body ambiguity: the under-determination of the raw multibody impulse exchange discussed in the main development is precisely the absence of the off-diagonal blocks of M_a and their configuration gradient. Forming $\mathsf{M}(q)$ makes the exchange explicit.

2 Explicit assembly of the added-mass metric

The influence matrix A of eq. (5) depends only on geometry, not on $\mathbf{z}$, and its factorisation is already cached (§7). Column j of M_a is obtained by imposing a unit generalized velocity $\mathbf{z} = \mathbf{e}_j$ (unit translation or rotation of one body about one axis), solving the *same* BEM system for the resulting ϕ, and reading the stacked impulse (3). All $6N$ right-hand sides share the cached factorisation, so the entire matrix costs one factorisation and $6N$ back-substitutions/warm-started GMRES solves rather than $6N$ independent solves. Symmetrising $\frac{1}{2}(\mathsf{M}_a + \mathsf{M}_a^\top)$ removes the small discretisation asymmetry.

This assembly was verified directly. At a close two-body configuration (ndiv = 2) the assembled M_a is symmetric to 3×10^{-5} (relative), reproduces the solver's fluid kinetic energy through (4) to 6×10^{-4}, and its configuration gradient contracted with the velocity, $\frac{1}{2}\,\mathbf{z}^\top(\partial_q \mathsf{M}_a)\mathbf{z}$, matches the finite-differenced $\partial_q T_f$ to 2×10^{-5}. This last identity is the body–body exchange force that the impulse scheme's pair-metric correction approximates, here obtained exactly. The full matrix costs about $1.9\times$ a single impulse solve, confirming the shared-factorisation economics.

3 Body-frame Kirchhoff equations

Working in the lab frame at fixed $\mathbf{z}$ and differentiating T with respect to orientation double-counts the kinematic link between R_i and $\boldsymbol{\omega}_i$, injecting a spurious $-\boldsymbol{\omega}_i \times \mathbf{H}_i$ torque for anisotropic bodies. We therefore pass to *body-frame* velocities and momenta, in which each body's self block is constant. For body i let

$$\mathbf{V}_i = R_i^\top \mathbf{v}_i, \quad \boldsymbol{\Omega}_i = R_i^\top \boldsymbol{\omega}_i, \qquad \zeta = (\mathbf{V}_1, \boldsymbol{\Omega}_1, \dots), \qquad \mathsf{M}_{\mathrm{bf}}(q) = \mathsf{T}(R)^\top\, \mathsf{M}(q)\, \mathsf{T}(R), \tag{6}$$

with $\mathsf{T}(R) = \mathrm{blkdiag}(R_i, R_i)$, and body-frame momenta $\boldsymbol{\mu} = \mathsf{M}_{\mathrm{bf}}\zeta$, written per body as $(\mathbf{P}_i, \boldsymbol{\Pi}_i)$. The reduced Euler–Poincaré (Kirchhoff) equations on $\mathrm{SE}(3)^N$ are

$$\dot{\mathbf{P}}_i = \mathbf{P}_i \times \boldsymbol{\Omega}_i + \mathbf{F}_i, \tag{7}$$

$$\dot{\boldsymbol{\Pi}}_i = \boldsymbol{\Pi}_i \times \boldsymbol{\Omega}_i + \mathbf{P}_i \times \mathbf{V}_i + \mathbf{T}_i, \tag{8}$$

where $(\mathbf{F}_i, \mathbf{T}_i)$ is the body-frame configuration force, i.e. the body-referred gradient of the interaction energy at fixed ζ. For a single body M_{bf} is constant, $(\mathbf{F}, \mathbf{T}) = \mathbf{0}$, and (7)–(8) reduce to the classical Kirchhoff top; the gyroscopic terms $\mathbf{P} \times \boldsymbol{\Omega}$ and $\boldsymbol{\Pi} \times \boldsymbol{\Omega} + \mathbf{P} \times \mathbf{V}$ are exactly what the naive lab-frame gradient omitted.

Configuration force and the momentum identities

The interaction energy is invariant under a rigid motion of the whole configuration at fixed body-frame ζ; hence the true configuration force is L^2-orthogonal to the six global rigid-motion directions of Q, which expresses the discrete Noether identities

$$\sum_i R_i \mathbf{F}_i = \mathbf{0}, \qquad \sum_i \big(\mathbf{c}_i \times R_i \mathbf{F}_i + R_i \mathbf{T}_i\big) = \mathbf{0}. \tag{9}$$

A finite-difference estimate of $(\mathbf{F}, \mathbf{T})$ violates (9) only at truncation order; projecting it onto the complement of the six rigid-motion directions restores both identities exactly and is essential for machine-precision momentum conservation.

4 Structure-preserving time integration

Write $g = (R, \mathbf{c}) \in \mathrm{SE}(3)$ per body and let $\boldsymbol{\mu} = (\mathbf{P}, \mathbf{\Pi})$ transform under the coadjoint action of $f = (R_f, \mathbf{p}_f) \in \mathrm{SE}(3)$ as

$$\mathrm{Ad}_f^* \boldsymbol{\mu} = \big(R_f^\top \mathbf{P},\ R_f^\top (\mathbf{\Pi} + \mathbf{P} \times \mathbf{p}_f)\big), \tag{10}$$

whose infinitesimal form is the right-hand side of (7)–(8). The spatial momentum $\mathrm{Ad}_{g^{-1}}^* \boldsymbol{\mu}$ is the conserved Noether quantity. One implicit-midpoint step, with unknown midpoint velocity ζ_{mid}, is

$$\zeta_{\mathrm{mid}} = \mathsf{M}_{\mathrm{bf}}(q_{\mathrm{mid}})^{-1}\Big(\mathrm{Ad}^*_{f_{n\to\mathrm{mid}}}\boldsymbol{\mu}_n + \tfrac{1}{2}\,\Delta t\,\mathbf{f}_{\mathrm{cfg}}\Big), \tag{11}$$

$$q_{n+1} = q_n \oplus \Delta t\,\zeta_{\mathrm{mid}} \quad (\mathbf{c}_i{+}{=}\Delta t\, R_{\mathrm{mid},i}\mathbf{V}_i,\ R_i \leftarrow R_i \exp(\Delta t\,\widehat{\mathbf{\Omega}}_i)), \tag{12}$$

$$\boldsymbol{\mu}_{n+1} = \mathrm{Ad}^*_{f_{\mathrm{mid}\to n+1}}\Big(\mathrm{Ad}^*_{f_{n\to\mathrm{mid}}}\boldsymbol{\mu}_n + \Delta t\,\mathbf{f}_{\mathrm{cfg}}\Big). \tag{13}$$

Here q_{mid} is the half-configuration, $\mathbf{f}_{\mathrm{cfg}}$ the projected configuration force there, and each leg increment $f = (R_a^\top R_b,\ R_a^\top(\mathbf{c}_b - \mathbf{c}_a))$ is the *exact* relative motion between the two configurations it connects, so the two legs $n \to \mathrm{mid} \to n{+}1$ in (13) compose to the exact full increment $n \to n{+}1$. Consequently

$$\mathrm{Ad}^*_{g_{n+1}^{-1}}\boldsymbol{\mu}_{n+1} = \mathrm{Ad}^*_{g_n^{-1}}\boldsymbol{\mu}_n + \Delta t \sum_i (\text{spatial } \mathbf{F}_i), \tag{14}$$

and the sum vanishes by (9): **the total linear and angular spatial momentum are conserved to machine precision**. Two points make this exact rather than $O(\Delta t^2)$: (i) the impulse is applied *once*, at the half configuration where the projection is performed, so its transported net force cancels; splitting it across the two legs would leave a term in the $n{+}1$ frame and leak momentum at $O(\Delta t^2)$ per step; and (ii) the leg increments are the true relative motions, not the exponential of an averaged twist.

Energy is not conserved exactly—as for any momentum-conserving variational integrator—but is bounded with no secular drift. For the free Kirchhoff top, (11) with $\mathbf{f}_{\mathrm{cfg}} = \mathbf{0}$ conserves both the energy $\frac{1}{2}\zeta^\top \mathsf{M}_{\mathrm{bf}}\zeta$ and the Casimir $|\mathbf{\Pi}|$ to machine precision.

Predictor form for efficiency

Evaluating M_{bf} and $\mathbf{f}_{\mathrm{cfg}}$ inside the fixed point makes each iterate a full metric assembly. Because the metric at the true versus predicted midpoint differ by $O(\Delta t^2)$, we instead assemble both *once* per step at a midpoint predicted from the previous step's ζ_{mid}, and take a single direct (BEM-free) update (11)–(13). Carrying $\boldsymbol{\mu}$ as the state removes any endpoint metric solve, and the configuration force uses forward differences. Crucially the *same* predicted midpoint is used for the projection and for the transport intermediate in (13), so momentum remains machine-exact while second order is preserved.

5 Properties and verification

Table 1 summarises the behaviour on a spinning single ellipsoid and a close two-body pair ($1{:}0.8{:}0.6$ ellipsoids, sep 3.5, ndiv $= 2$, $\Delta t = 0.05$). Momentum is conserved to $\sim 10^{-13}$; energy is bounded; the trajectory is second order (error $\propto \Delta t^2$ against a fine reference). Compared against the production variational scheme on the same close pair, both integrators converge to the same continuum trajectory, and the reduced-metric scheme is about an order of magnitude more accurate at a given step, while its predictor form runs at a small multiple of the impulse scheme's per-step cost.

case / scheme	energy drift	$\lvert\Delta\mathbf{P}\rvert$	$\lvert\Delta\mathbf{H}\rvert$
single ellipsoid, free top	9×10^{-14}	2×10^{-14}	3×10^{-14}
two ellipsoids, close pair	8×10^{-5} (bounded)	9×10^{-14}	1×10^{-13}
trajectory error vs. fine ref. (Δt=0.05)	1.1×10^{-4} (2nd order)		
production variational, same case	5.5×10^{-3} (1st order)		
cost: impulse / reduced-metric (once-per-step)	0.12 / 0.29 s per step		

Table 1: Reduced-metric integrator: conservation, accuracy and cost. Momentum drifts are the maxima over the run; energy drift is relative.

Relation to the impulse scheme. The impulse scheme evaluates only $\boldsymbol{\ell}(q,\mathbf{z})$ (one BEM solve) and differences it; it never forms M_a or $\partial_q\mathsf{M}_a$, which is both why it is fast and why its multibody exchange is under-determined. The reduced-metric scheme pays for the metric and its gradient and, in return, is fully determined and momentum-exact by construction. The remaining cost gap is entirely the finite-difference configuration force (a geometry re-mesh per perturbation); replacing it with an analytic added-mass shape derivative (Hadamard/adjoint) is the route to full parity and is the natural next development.